

Q1 - Ideal gas: It takes 80 cubic feet of air at atmospheric pressure to fill a SCUBA tank. The pressure after filling is 200 times atmospheric. A) Roughly how much more does a full tank weigh than an empty one (please answer in kg)? B) What is the volume in litres of the interior of a tank? C) (challenge, would not be on a quiz): How much work did we do filling the tank? SCUBA tanks are always placed in water when filled so all that work has somewhere to go. Otherwise the tanks would seriously overheat.

A) $PV=nkT$. At atmospheric pressure, the pressure is 10^5 pascals, the volume is 80 cubic feet, and the temperature is around 300K. We first need to convert the volume to SI. There are 3.3 feet per metre, so one cubic metre is $3.3^3 = 36$ cubic feet. We have 80 cubic feet to start with, so we have $80/36 = 2.2m^3$. You could either remember that the density of air is about $1 \text{ kg}/m^3$, or we could work it out. We have $n = PV/kT = 10^5 * 2.2 / (1.38e-23 * 300) = 5.3e25$. If we divide by Avogadro's number, we know we have 90 moles of air, which weighs around 29 grams per mole (nitrogen atoms have 7 protons and 7 neutrons for 14 AMU, so a nitrogen molecule is 28 AMU. Oxygen is a bit more, so we'll take 29 for the average, which means a mole of air is about 29 grams). $90*29=2610$ grams, or 2.6 kg.

B) We worked out already that the volume of air before we compressed it was 2.2 cubic metres. The pressure went up by a factor of 200 while the temperature stayed fixed, so the volume went down by 200. That means the volume is $2.2 \text{ metres}^3 / 200 = 0.011 \text{ metres}^3$ (the actual answer is 12 for a standard tank).

C) This *could* be a tricky question, because if we let the tank heat up while we fill it, we have to do more work to get the same amount of gas in. Because tanks are held in a tank of water, though, that doesn't happen. The work to squeeze a bit of gas into a volume is the pressure. More completely, $work = PdV + VdP$ but since the tank has fixed volume, we only worry about VdP . We start at 10^5 pascals but end up at 200 times that, or $2e7$ pascals. The volume is around 0.011 m^3 , so we do a total of $2e7 * 1.1e-2 = 2.2e5$ Joules. If you do a more careful calculation that requires calculus, you get an actual answer about 5 times higher so we're in the ballpark - essentially, we got the right answer for how much *net* work we did, but we had to do some extra work that goes into heating up the air that then goes into the water. If all that work stayed in the scuba tank, the temperature would go up by 50-100 degrees - much too hot to touch!

Q2 - Units: Newton's law of gravitation says that $F = GM_1M_2/R^2$, where F is a force, G is a constant, M_1 and M_2 are masses, and R is a distance. A) What are the units of G ? please express using only kg, metres, and seconds. B) A cloud of gas has radius R and density ρ (units kg/m^3). About how long would it take to collapse? Hint - take G , the density, and the radius and try to make a time. You may or may not need all the information I have given you. C) (challenge, would never be on quiz/test): the apparent diameter of the sun is about 0.01 radians, and it takes a year for the Earth to go around the sun. What is the average density of the sun?

A) We know $F = GM_1M_2/R^2$ or $G = \frac{FR^2}{M_1M_2}$. If we plug in units, we have units of G are Nm^2/kg^2 . If we let a Newton be a kgm/s^2 , we have $G = kgm/s^2 * m^2/kg^2 = \frac{m^3}{s^2kg}$.

B) We know we need to use G to get a time. We know we don't need a kg to get a time, and the only other thing with units of kg we have is the density. Density has units of kg/m^3 while G has one copy of kg in the denominator, so $G\rho$ will get rid of our density. What happens now? The units of $G\rho$ are $\frac{m^3}{s^2 kg} \frac{kg}{m^3} = \frac{1}{s^2}$. Getting rid of the kg in the units also got rid of our size. If we have $G\rho = \frac{1}{s^2}$ then a quantity with seconds will be $\frac{1}{\sqrt{G\rho}}$. No matter how big our cloud is, a collapse will take the same amount of time if it has the same density. That would be hard to show with physics, but units lets us figure it out with (almost) no work!

The Earth takes a year to go around the sun, so the length of time to fall into the sun has to be similar. It's probably less, so we'll round down. Useful fact (for some definitions of useful) is that there are about $\pi e7$ seconds in a year, so let's guess it's about $1e7$ seconds to fall in. G is $6.67e-11$, so the average density inside of the Earth's orbit must be something like:

$$t = \frac{1}{\sqrt{G\rho}} \quad (1)$$

$$t^2 = \frac{1}{G\rho} \quad (2)$$

$$\rho = \frac{1}{Gt^2} \quad (3)$$

if we plug in numbers, we have $\rho = \frac{1}{6.67e-11 \times 1e14} = 1.4e-4 kg/m^3$. The diameter of the sun is 0.01 radians, so the radius must be $1/200$ radians. That means the radius of the sun is $1/200$ the radius of the Earth's orbit, so the sun fills a *fraction* of the volume of $1/200^3$. Since (almost) all the mass is in the sun, the average density of just the sun must be a factor of 200^3 higher than the average density. We get $\rho_{sun} = 1.4e-4 * 200^3 = 1200 kg/m^3$, or in more familiar (but not SI!) units, 1.2 grams per cubic centimetre. The actual answer is 1.4, so we're very close!

Q3 - sin/cos: Half angle formula. A) Express $\cos(2\theta)$ only in terms of $\cos(\theta)$ and constants (i.e. eliminate any $\sin(\theta)$ terms). Hint, $2\theta = \theta + \theta$. B) Express $\cos(\theta)$ only in terms of $\cos(2\theta)$ and constants. And don't forget that $\sin^2 + \cos^2 = 1$.

A) we know $\cos(2\theta) = \cos(\theta + \theta) = \cos^2(\theta) - \sin^2(\theta)$. We want to get rid of $\sin^2$, but we know that since $\sin^2 + \cos^2 = 1$ that $\sin^2 = 1 - \cos^2$. If we plug that in, we know that

$$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta) = \cos^2(\theta) - (1 - \cos^2(\theta)) = 2\cos^2\theta - 1 \quad (4)$$

B) We take the previous answer and now solve for $\cos(\theta)$:

$$\cos(2\theta) = 2\cos(\theta)^2 - 1 \quad (5)$$

$$2\cos(\theta)^2 = 1 + \cos(2\theta) \quad (6)$$

$$\cos(\theta)^2 = \frac{1 + \cos(2\theta)}{2} \quad (7)$$

$$\cos(\theta) = \sqrt{\frac{1 + \cos(2\theta)}{2}} \quad (8)$$

If I say let $\phi = 2\theta$, then this becomes

$$\cos(\phi/2) = \sqrt{\frac{1 + \cos(\phi)}{2}} \quad (9)$$

Equation 9 tells us that if we know the cosine of an angle, we can also work out the cosine of half of that angle.

Possibly useful quantities: proton mass is $1/(6.e26)$ kg. There are about 30 litres in a cubic foot. $k_B = 1.38e - 23$ Joules/Kelvin.